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Deep Learning for NLP

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1. Abstract

This report presents the development and evaluation of classifiers for response clarity classification in political question–answer (QA) pairs. The task uses the dataset released for SemEval 2026 Task 6 (CLARITY) and requires predicting whether a president's answer to a journalist's question is a Clear Reply, an Ambivalent response, or a Clear Non-Reply.

To tackle this three class classification problem three pretrained transformer based models are fine tuned: BERT (bert-base-uncased), DistilBERT (distilbert-base-uncased), and DeBERTa-v3-base. All models are implemented in PyTorch with a custom explicit training loop.

Prior to model training the dataset undergoes data cleaning. For all three models systematic hyperparameter optimization is performed using optuna. Models are evaluated and compared using weighted F1 score as the primary metric with Macro F1 score reported as an additional metric.

2. Data processing and analysis

2.1. Data Cleaning

The first cleaning step involves identifying and removing (question, answer) pairs with conflicting clarity labels. Since the dataset was annotated by multiple annotators the same (question, answer) pair could receive different clarity labels from different annotators. Keeping such pairs would introduce noise into the training process, so for the same (question, answer) pair if the annotators did not agree on the same clarity label this pair is being dropped. This cleaning step is applied only to the train set where 12 conflicting pairs were found as the test set contains no conflicting pairs.

The second cleaning step removes rows marked as inaudible (inaudible = True). This means that during the interview the audio quality was poor or unclear at that specific point so the annotator could not hear the answer properly making these entries useless for training the classifier. In the train set 45 such rows were found and dropped.

2.2. Analysis

The exploratory data analysis is performed on the train set and on the test set after the data cleaning steps. The train set contains 3,379 samples and the test set contains 308 samples covering interviews from four U.S. presidents: Donald J. Trump, Barack Obama, George W. Bush and Joseph R. Biden.

Label Distribution
The dataset is significantly imbalanced across all three classes. In the train set, which contains 3,379 samples the Ambivalent class dominates with 2,001 samples (59.2%), followed by Clear Reply with 1,029 samples (30.5%) and Clear Non-Reply with only 349 samples (10.3%). The same imbalance pattern is observed in the test set, which contains 308 samples where Ambivalent accounts for 206 samples (66.9%), Clear Reply

for 79 samples (25.6%) and Clear Non-Reply for only 23 samples (7.47%). This class imbalance is an important consideration for model training as a classifier that simply predicts Ambivalent for every sample would already achieve a high accuracy. This is why weighted F1-score is used as the primary evaluation metric instead of accuracy, as it accounts for class imbalance by weighting each class proportionally to its support.

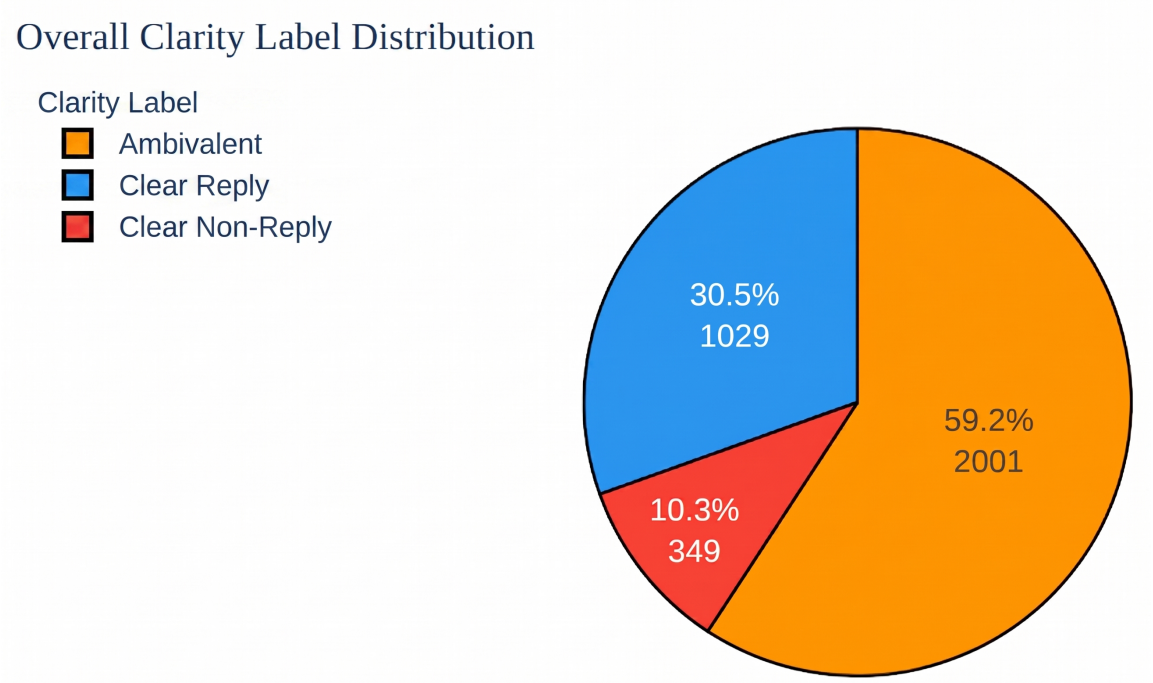


Figure 1: Label distribution in the training set

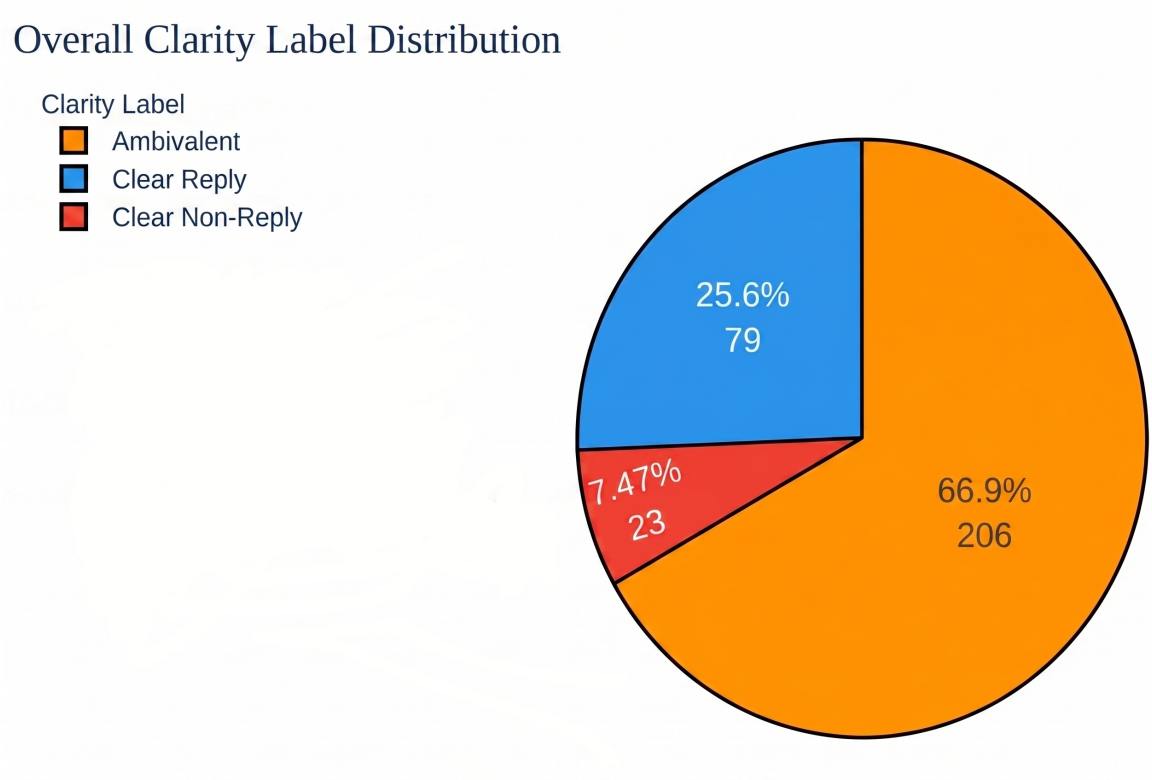


Figure 2: Label distribution in the test set

Answer Length Statistics
The answer length statistics reveal meaningful differences across the three clarity labels. Ambivalent answers are on average the longest with a mean of 327.9 words and a median of 241.0 words suggesting that when presidents give ambivalent responses they tend to speak at length without fully committing to a clear answer. Clear Reply answers have an average of 267.9 words and a median of 190.0 words. Clear Non-Reply answers are notably shorter with an average of only 136.0 words and a very low median of 40.0 words indicating that when presidents choose not to answer a question they tend to do so with very few words rather than giving a long evasive response.

Label	Average Length	Median	Min	Max
Clear Reply	267.9	190.0	1	1592
Ambivalent	327.9	241.0	1	2117
Clear Non-Reply	136.0	40.0	1	1407

Table 1: Answer length statistics per label on the training set

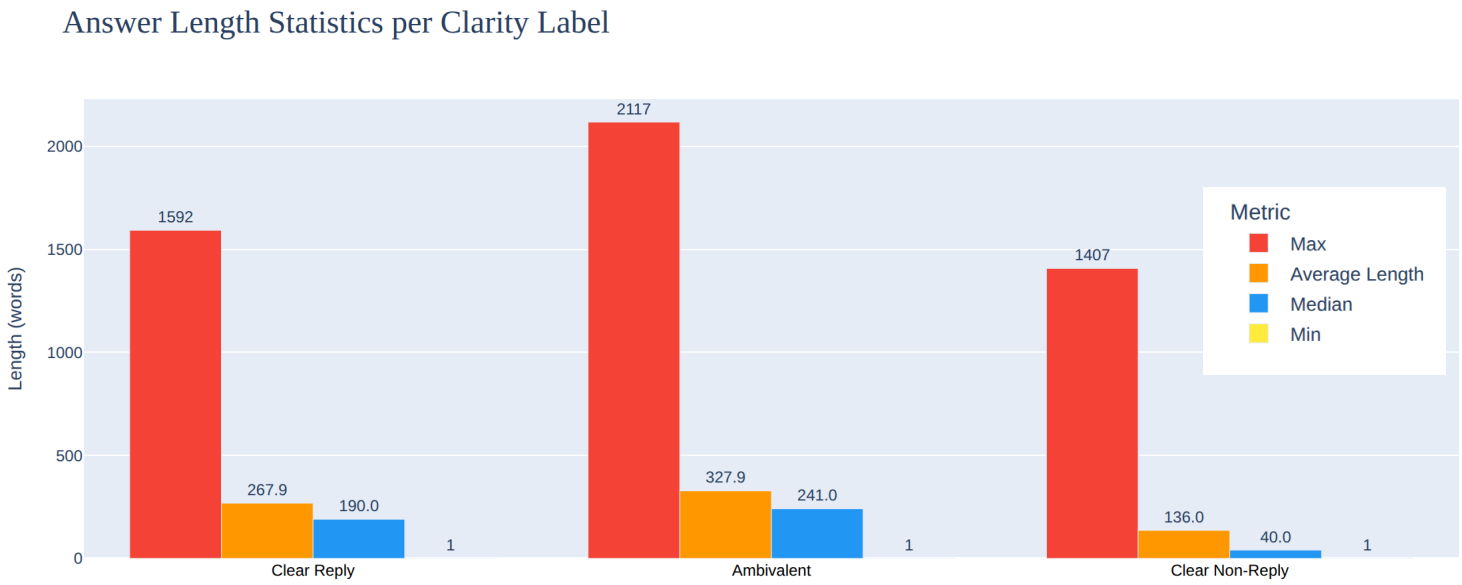


Figure 3: Answer length distribution by clarity label in the training set

2.3. Data partitioning for train, test and validation

The dataset was obtained from HuggingFace and comes pre-divided into two splits: a train split and a test split.

The train split was further divided into a training set (85%) and a validation set (15%) using scikit-learn train_test_split function. The validation set was used to evaluate the model's performance during development without touching the test set.

Two important choices were made during the split:

- Stratification: The stratify=labels parameter was used to ensure that the class distribution of the three clarity labels (Clear Reply, Ambivalent, Clear Non-Reply) is preserved proportionally in both the training and validation sets. This is particularly important since the dataset is imbalanced, with the Ambivalent class accounting for approximately 59.2% of the training data. Stratification ensures that all three classes (Clear Reply, Ambivalent and Clear Non-Reply) are adequately represented in both the training and validation sets.
- Random seed: A fixed random_state=42 was used to ensure reproducibility of the results.

The 85/15 ratio was chosen after experimental evaluation. During our experiments we tested a 90/10 split to provide the model with more training data. However, this consistently hurt performance across both models for BERT, the test F1 (weighted) dropped from 0.6500 to 0.5538 and the Clear Reply class nearly collapsed with a recall of only 0.01, while for DistilBERT the test F1 dropped from 0.6219 to 0.6149.

2.4. Input Representation

Each sample in the dataset consists of two text fields: a question asked by a journalist and an answer given by a president. Since transformer models expect a single input sequence, the question and answer must be combined into a single input. We adopt the standard two segment input format where the question and answer are concatenated using special tokens in the following format:

[CLS] question [SEP] answer [SEP]

The [CLS] token is placed at the beginning of the sequence and its final hidden state is used by the classification head to predict the clarity label. The [SEP] tokens act as separators, explicitly marking the boundary between the question and the answer.

A key challenge with this dataset is that many combined question-answer sequences are very long and exceed the maximum sequence length of 512 tokens supported by these models. Analysis of the training set revealed that when the question and answer are concatenated in the format [CLS] question [SEP] answer [SEP], approximately 30% of the dataset exceeds the 512 token limit. To handle this, a custom truncation strategy was designed. The question is capped at a maximum of 130 tokens because from the dataset we saw that the maximum question length is approximately 100 tokens. If the question is shorter than 130 tokens, the remaining space is dynamically allocated to the answer, maximizing the amount of answer content that fits within the 512 token limit. This dynamic allocation is applied consistently across all three truncation strategies.

Three truncation strategies were explored for handling long answers: head only, tail only and head + tail. In the head + tail strategy the remaining available space is split equally between the beginning and the end of the answer. Our experiments confirmed that the head + tail strategy consistently achieved the best results compared to head only and tail only strategies, which suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity, as will be discussed in detail in Section 3.4. All sequences are padded to exactly 512 tokens, with attention masks used to distinguish real tokens from padding tokens. The same input representation and truncation strategy was applied consistently across all three models to ensure a fair comparison.

3. Algorithms and Experiments

3.1. Models

BERT (bert-base-uncased)

BERT (Bidirectional Encoder Representations from Transformers)is a pretrained transformer based language model. The bert-base-uncased variant consists of 12 transformer layers, 12 attention heads, a hidden size of 768 and approximately 110 million parameters.

What makes BERT unique is its bidirectional nature, i.e. it processes the entire input sequence simultaneously in both directions, allowing each token to attend to all other tokens in the sequence. BERT was pretrained on BookCorpus and English Wikipedia

using two objectives: Masked Language Modeling (MLM), where random tokens are masked and the model learns to predict them from context and Next Sentence Prediction (NSP), where the model learns to predict whether two sentences follow each other in the original text.

BERT is well suited for this task because determining whether a political answer is clear, ambivalent, or evasive requires understanding the full context of both the question and the answer together, which bidirectional attention captures effectively by allowing every token to attend to every other token in the input. For classification, we fine tune BertForSequenceClassification, which adds a linear classification head on top of the [CLS] token representation. The [CLS] token aggregates the contextual information of the entire input sequence and its final hidden state is fed into the classification head to produce the three class scores.

DistilBERT (distilbert-base-uncased)

DistilBERT is a smaller and faster version of BERT. The distilbert-base-uncased variant consists of 6 transformer layers, 12 attention heads, a hidden size of 768 and approximately 66 million parameters making it 40% smaller and 60% faster than BERT.

What makes DistilBERT unique is how it was created. Instead of being pretrained from scratch DistilBERT is trained through knowledge distillation, a process where a smaller student model learns to mimic the behavior of a larger teacher model in this case BERT. This allows DistilBERT to retain approximately 97% of BERT's performance on standard benchmarks while being significantly more efficient in terms of both memory and computation.

Since DistilBERT inherits BERT's core architecture and bidirectional attention mechanism through knowledge distillation, it retains the same ability to capture contextual relationships between the question and the answer. The key difference is that it does so with fewer parameters and faster inference making it a useful baseline for investigating whether a smaller and more efficient model can achieve comparable performance to BERT on this task. For classification, we fine tune DistilBertForSequenceClassification in the same way as BERT, adding a linear classification head on top of the [CLS] token representation.

DeBERTa-v3-base (microsoft/deberta-v3-base)

DeBERTA (Decoding-enhanced BERT with Disentangled Attention) is a pretrained transformer based language model developed by Microsoft. The deberta-v3-base variant consists of 12 transformer layers, 12 attention heads, a hidden size of 768 and approximately 184 million parameters, making it significantly larger than BERT (110M).

What makes DeBERTA unique is its disentangled attention mechanism, which represents each token using two separate vectors, one for its content and one for its position and computes attention weights using disentangled matrices for content-to-content, content-to-position, and position-to-content interactions. This allows the model to better capture the relationship between the meaning of a word and its position in the sequence compared to standard BERT-style attention. DeBERTa-v3 was pretrained on Wikipedia, BookCorpus and additional web crawl data using Replaced Token Detection (RTD), where the model learns to distinguish real tokens from tokens replaced

by a generator model, which provides a richer training signal than BERT's Masked Language Modeling.

DeBERTA is well suited for this task because its disentangled attention mechanism allows it to better capture the complex relationship between the journalist question and the presidential answer, which is fundamental for determining the clarity label of a political response. For classification, we fine tune AutoModelForSequenceClassification with DeBERTa-v3-base as the backbone, which adds a linear classification head on top of the model output to produce the three class scores.

3.2. Training Setup and Optimization

All three models were trained using the same optimization setup

Optimizer
The AdamW optimizer was used for training both models. AdamW is a variant of the Adam optimizer that decouples weight decay from the gradient update, making regularization more effective. It is the standard optimizer recommended for fine tuning transformer based models.

Learning Rate Scheduler
A linear learning rate scheduler was used, which gradually decreases the learning rate from its initial value to zero over the total number of training steps. This allows the model to make larger updates at the beginning of training and increasingly smaller and more precise updates as training progresses, which helps stabilize convergence in the later epochs.

Gradient Clipping
Gradient clipping with a maximum norm of 1.0 was applied after each backward pass. This prevents the gradients from becoming too large during backpropagation, which could cause unstable weight updates and break training. A clipping value of 1.0 is the standard value used in the original BERT paper.

Class Weights
The dataset is significantly imbalanced across the three classes. To address this a class-weighted loss function was used during training. Class weights were computed using scikit-learn's compute_class_weight function with the balanced setting, which assigns higher weights to minority classes and lower weights to the dominant Ambivalent class. This encourages the model to pay more attention to underrepresented classes during training.

Reproducibility
A fixed random seed of 42 was set across all random number generators to ensure that all experiments are fully reproducible.

3.3. Hyperparameter Tuning with optuna

Hyperparameter tuning was performed using optuna, an automatic hyperparameter optimization framework that systematically searches for the best combination of training parameters. Instead of manually trying different configurations, optuna runs

multiple trials where each trial trains the model with a different set of hyperparameters and evaluates it on the validation set. This approach is clearly more efficient than manual tuning and explores a wider range of configurations.

Each trial starts with fresh pretrained weights to ensure that results are not influenced by previous trials. To avoid wasting time on weak configurations, the MedianPruner was used because it stops a trial early if its validation F1 at a given epoch is worse than the median of all previous trials at the same epoch. The best trial is selected based on the highest validation F1 (weighted) achieved across all epochs and all trials.

The parameters searched and their ranges are shown in Table 2. Note that the search space evolved across experiments

Parameter	**Search Range**
Learning Rate	[1e-5, 3e-5]
Weight Decay	[0.0, 0.1]
Epochs	[2, 5]
Batch Size	{8, 16, 32}
Warmup Ratio	[0.0, 0.1]

Table 2: Hyperparameter search space used across experiments

3.4. Experiments

3.4.1. Table of trials BERT Model Part.

BERT Experiments — Ranked by Test F1 (weighted)

Rank	Exp	Key Changes	Best Parameters	Truncation	Val F1 (weighted)	Test F1 (weighted)	Test Macro F1
1	Exp 4	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [2,5], wd ∈ [0.0, 0.01], 85/15 split	lr=2.99e-05, wd=0.0093, epochs=4	head + tail	0.6694	0.6500	0.5809
2	Exp 8	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [2,5], wd ∈ [0.0, 0.01], stopword removal, 85/15 split	lr=2.48e-05, wd=0.0035, epochs=4	head + tail	0.6549	0.6391	0.5311
3	Exp 6	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [2,5], wd ∈ [0.0, 0.01], 85/15 split	lr=2.53e-05, wd=0.0083, epochs=5	head only	0.6597	0.5939	0.4935
4	Exp 5	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [2,5], wd ∈ [0.0, 0.01], warmup ∈ [0.0, 0.1], n_trials=20, 85/15 split	lr=2.06e-05, wd=0.0077, warmup=0.0728, epochs=5	head + tail	0.6550	0.5930	0.5080
5	Exp 9	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [2,5], wd ∈ [0.0, 0.01], 85/15 split	lr=2.64e-05, wd=0.0089, epochs=5	tail only	0.6361	0.5880	0.4220
6	Exp 3	lr ∈ [1e-5, 3e-5], batch ∈ {16,32}, epochs ∈ [2,5], no weight decay, 85/15 split	lr=2.93e-05, batch=16, epochs=5	head + tail	0.6551	0.5636	0.4364
7	Exp 7	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [2,5], wd ∈ [0.0, 0.01], 90/10 split	lr=1.21e-05, wd=0.0073, epochs=4	head + tail	0.6265	0.5538	0.3565

wd == weight decay

3.4.2. BERT Experiments.

All experiments used the head + tail truncation strategy unless stated otherwise, as shown by Experiments 6 and 9, which instead used head only or tail only truncation and consistently achieved worse results, which suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity.

Note that Experiments 1 and 2 were initial runs used only to verify that the code was working correctly with fixed hyperparameters. Since they were not part of the systematic search and their results are not meaningful for comparison they are not reported here. The systematic experimentation starts from Experiment 3.

Experiment 3 — optuna search with Batch Size Exploration
The first systematic experiment introduced optuna hyperparameter search over learning rate, batch size and number of epochs across 15 trials. The best trial achieved a validation F1 of 0.6551 but the test F1 dropped significantly to 0.5636, revealing a large gap between validation and test performance. This gap indicates overfitting the model learned to fit the validation set well but failed to generalize to unseen data. Additionally batch size 32 consistently underperformed compared to batch size 16 across all trials, suggesting that smaller batches provide better gradient estimates for this task. Weight decay was not part of the search space in this experiment, which is the main reason for the overfitting observed.

Experiment 4 — Adding weight decay with head + tail truncation strategy (Best Experiment)
Based on the overfitting observed in Experiment 3 weight decay was added to the optuna search space and batch size was fixed to 16. This was the most impactful change across all experiments, producing not only the best test F1 (weighted) of 0.6500 but also the best validation F1 (weighted) of 0.6694 across all experiments. Critically, the gap between validation F1 weighted (0.6694) and test F1 weighted (0.6500) is the smallest across all experiments, confirming that the model generalized well to unseen data and was not overfitting unlike Experiment 3 where the same gap was much larger (0.6551 vs 0.5636). Weight decay acts as a regularizer by penalizing large weights, which prevents the model from memorizing the training data and improves generalization to unseen samples. This confirms that regularization is essential when fine tuning large pretrained models on relatively small datasets. Additionally this experiment used the head + tail truncation strategy, which as confirmed by Experiments 6 and 9 which instead used head only or tail only truncation and consistently achieved worse results. This suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity and that by preserving both parts we capture more of the information needed to determine the clarity label of a political answer. The combination of weight decay regularization and head + tail truncation makes this the best overall configuration for BERT on this task.

Experiment 5 — Adding Warmup Ratio
Warmup ratio was added to the optuna search space to investigate whether gradually increasing the learning rate at the start of training could stabilize early updates and improve performance. Additionally the number of trials was increased to 20. However test F1 (weighted) dropped from 0.6500 to 0.5930 compared to Experiment 4. Although the validation F1 (weighted) remained similar (0.6550), the gap between validation and test performance widened again, suggesting that the warmup ratio introduced instability rather than helping generalization. At this scale BERT does not benefit from a gradual learning rate warmup and the linear scheduler alone is sufficient.

Experiment 6 — Head Only Truncation
The head + tail truncation strategy was replaced with head only truncation to investigate whether the beginning of a political answer is more informative than its end for clarity classification. Test F1 dropped from 0.6500 to 0.5939, confirming that discarding the tail of the answer hurts performance. This suggests that the conclusion of a political answer also carries important clarity signals, for example a politician may start ambiguously but give a clearer or more evasive conclusion at the end. By keeping only

the beginning of the answer the model loses access to this information.

Experiment 7 — Smaller Validation Split (90/10)
The validation split was reduced from 15% to 10% to give the model more training data. This produced the worst result across all experiments the validation F1 weighted dropped to 0.6265 compared to 0.6694 in Experiment 4, and the test F1 weighted dropped further to 0.5538. Most strikingly, Clear Reply recall collapsed to almost zero (0.01) (see Table 4), meaning the model almost never predicted this class.

Experiment 8 — Stopword Removal
A custom stopword removal step was added to the preprocessing pipeline removing common uninformative words. Although the validation F1 weighted (0.6549) was close to Experiment 4 (0.6694), the test F1 weighted dropped slightly to 0.6391 compared to Experiment 4 (0.6500). This is because BERT is a contextual model pretrained on full sentences including function words so removing stopwords before tokenization disrupts the natural language patterns BERT learned during pretraining and therefore slightly hurts performance. This confirms that manual preprocessing steps are generally not beneficial when fine tuning transformer models, as BERT already handles context and word importance internally through its attention mechanism.

Experiment 9 — Tail Only Truncation
The head + tail truncation strategy was replaced with tail only truncation to test whether the end of a political answer is more informative than its beginning. The test F1 weighted dropped to 0.5880 the worst truncation strategy of the three tested, confirming that the model failed to generalize to unseen data. Clear Reply nearly collapsed with a recall of only 0.06 (see Table 5), suggesting that the beginning of a clear reply contains strong clarity signals such as direct affirmatives or specific factual statements that are completely lost when only the tail is kept. Comparing all three truncation strategies head only (0.5939), tail only (0.5880), and head + tail (0.6500) confirms that the head + tail strategy is clearly superior, suggesting that in political interviews the most relevant information for determining clarity is distributed across both the beginning and the end of an answer and that preserving both parts leads to better classification performance.

Experiment 4 (Best)

Class	Precision	Recall	F1
Clear Non-Reply	0.54	0.61	0.57
Ambivalent	0.76	0.73	0.74
Clear Reply	0.41	0.44	0.43

Table 3: Per class results for Experiment 4 (best experiment)

Experiment 7 (90/10 split)

Class	Precision	Recall	F1
Clear Non-Reply	0.22	0.30	0.25
Ambivalent	0.69	0.92	0.79
Clear Reply	1.00	0.01	0.03

Table 4: Per class results for Experiment 7 (90/10 split)

Experiment 9 (Tail only truncation)

Class	Precision	Recall	F1
Clear Non-Reply	0.28	0.48	0.35
Ambivalent	0.71	0.90	0.79
Clear Reply	0.71	0.06	0.12

Table 5: Per class results for Experiment 9 (tail only truncation)

3.4.3. Table of trials DistilBERT Model Part.

DistilBERT Experiments — Ranked by Test F1 (weighted)

Rank	Exp	Key Change	Best Parameters	Truncation	Val F1 (weighted)	Test F1 (weighted)	Test Macro F1
1	Exp 3	lr ∈ [1e-5, 2e-5], batch=16, epochs ∈ [3,5], wd ∈ [0.01, 0.1], warmup ∈ [0.0, 0.1], 85/15 split	lr=1.95e-05, wd=0.0571, warmup=0.0969, epochs=5	head + tail	0.6581	0.6219	0.5111
2	Exp 6	lr ∈ [1e-5, 2e-5], batch=16, epochs ∈ [3,5], wd ∈ [0.01, 0.1], warmup ∈ [0.0, 0.1], 90/10 split	lr=1.35e-05, wd=0.0786, warmup=0.0932, epochs=5	head + tail	0.6103	0.6149	0.5110
3	Exp 8	lr ∈ [1e-5, 2e-5], batch=16, epochs ∈ [2,4], wd ∈ [0.01, 0.1], no warmup, 85/15 split	lr=1.87e-05, wd=0.0693, epochs=4	tail only	0.6341	0.5949	0.4232
4	Exp 2	lr ∈ [1e-5, 2e-5], batch ∈ {16,32}, epochs ∈ [2,4], wd ∈ [0.01, 0.1], n_trials=20, 85/15 split	lr=1.65e-05, wd=0.0141, batch=16, epochs=4	head + tail	0.6195	0.5906	0.4971
5	Exp 5	lr ∈ [1e-5, 2e-5], batch=16, epochs ∈ [3,5], wd ∈ [0.01, 0.1], warmup ∈ [0.0, 0.1], 85/15 split	lr=1.95e-05, wd=0.0782, warmup=0.0881, epochs=5	head only	0.6356	0.5768	0.4541
6	Exp 1	lr ∈ [1e-5, 3e-5], batch=16, epochs ∈ [3,5], wd ∈ [0.0, 0.01], no warmup, 85/15 split	lr=1.48e-05, wd=0.0013, epochs=4	head + tail	0.6164	0.5583	0.4217
7	Exp 7	lr ∈ [1e-5, 2e-5], batch=16, epochs ∈ [3,5], wd ∈ [0.01, 0.1], warmup ∈ [0.0, 0.1], stopword removal, 85/15 split	lr=1.31e-05, wd=0.0882, warmup=0.0571, epochs=5	head + tail	0.6135	0.5540	0.3643

wd == weight decay

3.4.4. DistilBERT Experiments.

All experiments used the head + tail truncation strategy unless stated otherwise, as shown by Experiments 5 and 8, which instead used head only or tail only truncation and consistently achieved worse results, which suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity.

Experiment 1 — Baseline
The baseline experiment gave optuna the following parameters to search over: learning rate $\in$ [1e-5, 3e-5], batch size fixed to 16, epochs $\in [3, 5]$ and weight decay $\in [0.0, 0.01]$ with no warmup ratio across 15 trials. The best trial achieved a validation F1 weighted of 0.6164 and a test F1 weighted of 0.5583, suggesting that the low weight decay range was insufficient to prevent the model from overfitting to the dominant class.

Experiment 2 — Expanded Search Space
The optuna search space was expanded by narrowing the learning rate range to [1e-5, 2e-5], adding batch size 32 as an option, reducing epochs to [2,4], increasing weight decay to [0.01, 0.1] and increasing trials to 20. The best trial achieved a validation F1 weighted of 0.6195 and a test F1 weighted of 0.5906, an improvement over Experiment 1 (0.5583). Batch size 32 did not help the best trial selected by optuna used batch size 16, consistent with the finding from BERT experiments that smaller batches provide better gradient estimates for this task. A higher weight decay range helped the model generalize better, however the gap between validation F1 weighted (0.6195) and test F1 weighted (0.5906) remained large, suggesting that further improvements were needed.

Experiment 3 — Adding Warmup Ratio with head + tail Truncation Strategy (Best Experiment)
Warmup ratio was added to the optuna search space $\in [0.0, 0.1]$ and batch size was fixed back to 16 only. This was the most impactful change across all experiments, producing both the best test F1 weighted of 0.6219 and the best validation F1 weighted of 0.6581 across all experiments, confirming that the model generalized well to unseen data. The warmup ratio allows the learning rate to gradually increase from zero to its target value at the start of training, rather than starting with a large learning rate immediately. This prevents the model from making large unstable weight updates in the early epochs when the model weights are still far from a good solution and instead allows the model to settle into a stable learning trajectory before the learning rate reaches its full value, stabilizing early updates and helping the model better learn minority class patterns. Notably, Clear Non-Reply improved significantly from 0.31 in Experiment 1 to 0.50 in this experiment (see Table 7). This experiment also used the head + tail truncation strategy, which as confirmed by Experiments 5 and 8 which instead used head only or tail only truncation and consistently achieved worse results. This suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity and that by preserving both parts we capture more of the information needed to determine the clarity label of a political answer. The combination of warmup ratio, high weight decay and head + tail truncation makes this the best overall configuration for DistilBERT on this task. The high weight decay of 0.0571 acts as a strong regularizer by penalizing large weights, which prevents the model from memorizing the training data and improves generalization to unseen samples.

Experiment 5 — Head Only Truncation
The head + tail truncation strategy was replaced with head only truncation to investigate whether the beginning of a political answer is more informative than its end for clarity classification. The test F1 weighted dropped from 0.6219 to 0.5768, confirming that discarding the tail of the answer hurts performance and that the model did not

generalize well to unseen data. This suggests that the conclusion of a political answer also carries important clarity signals, for example a politician may start ambiguously but give a clearer or more evasive conclusion at the end. By keeping only the beginning of the answer the model loses access to this information.

Experiment 6 — Smaller Validation Split (90/10)
The validation split was reduced from 15% to 10% to give the model more training data. The validation F1 weighted dropped to 0.6103 compared to 0.6581 in Experiment 3 and the test F1 weighted dropped to 0.6149 compared to 0.6219 in Experiment 3, suggesting that having more training data did not lead to better performance in either validation or test F1.

Experiment 7 — Stopword Removal
A custom stopword removal step was added to the preprocessing pipeline removing common uninformative words. This produced the worst result across all experiments, with test F1 weighted dropping to 0.5540 compared to 0.6219 in Experiment 3. Most strikingly, Clear Reply completely collapsed, with a recall and F1 of 0.00 (see Table 8), meaning the model never predicted this class. This confirms that stopword removal severely disrupts the linguistic patterns DistilBERT learned during pretraining and therefore hurts performance. This confirms that manual preprocessing steps are generally not beneficial when fine tuning transformer models, as DistilBERT already handles context and word importance internally through its attention mechanism.

Experiment 8 — Tail Only Truncation
The head + tail truncation strategy was replaced with tail only truncation and warmup ratio was removed from the search space. The test F1 weighted dropped to 0.5949 compared to Experiment 3 (0.6219). Clear Reply nearly collapsed with a recall of only 0.08, suggesting that the beginning of a clear reply contains strong clarity signals that are completely lost when only the tail is kept. Comparing all three truncation strategies head only (0.5768), tail only (0.5949) and head + tail (0.6219) confirms that the head + tail strategy is clearly superior, suggesting that in political interviews the most relevant information for determining clarity is distributed across both the beginning and the end of an answer and that preserving both parts leads to better classification performance.

Experiment 1 (Baseline)

Class	Precision	Recall	F1
Clear Non-Reply	0.19	0.74	0.31
Ambivalent	0.73	0.67	0.70
Clear Reply	0.47	0.18	0.26

Table 6: Per class results for Experiment 1 (baseline)

Experiment 3 (Best)

Class	Precision	Recall	F1
Clear Non-Reply	0.42	0.61	0.50
Ambivalent	0.72	0.83	0.77
Clear Reply	0.41	0.19	0.26

Table 7: Per class results for Experiment 3 (best experiment)

Experiment 7 (Stopword Removal)

Class	Precision	Recall	F1
Clear Non-Reply	0.29	0.30	0.30
Ambivalent	0.69	0.94	0.80
Clear Reply	0.00	0.00	0.00

Table 8: Per class results for Experiment 7 (stopword removal)

3.4.5. Table of trials DeBERTA Model Part.

Note on Performance Gap Between Hardcoded Values vs Optuna Search Runs in DeBERTA : An interesting and unexpected observation was made during the DeBERTA experiments. When taking the best parameters found by the optuna search and running the model with those hardcoded values, a consistently large gap was observed between the test F1 weighted score of the hardcoded run and the original optuna run. This was tested multiple times across different runs with the optuna search, each time taking the best parameters found by optuna and running with those hardcoded values and every time the same pattern appeared i.e. the hardcoded run produced a noticeably lower test F1 weighted score compared to the optuna run that found those parameters. This behavior was not observed with BERT and DistilBERT, where running with hardcoded best parameters found by the optuna search consistently produced test F1 weighted scores very close to the original optuna results with only a small deviation.

Rank	Exp	Key Changes	Best Parameters	Truncation	Val F1 (weighted)	Test F1 (weighted)	Test Macro F1
1	Exp 1	lr ∈ [5e-6, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.1], warmup ∈ [0.0, 0.1], n_trials=5, 85/15 split	lr=1.78e-05, wd=0.0499, warmup=0.0794, epochs=3	head + tail	0.6823	0.6470	0.5551
2	Exp 9	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.06], warmup ∈ [0.06, 0.09], n_trials=7, 90/10 split	lr=1.86e-05, wd=0.0192, warmup=0.0697, epochs=5	head + tail	0.7067	0.6487	0.5464
3	Exp 8	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.06], warmup ∈ [0.06, 0.09], n_trials=7, 85/15 split	lr=1.69e-05, wd=0.0511, warmup=0.0775, epochs=5	tail only	0.6922	0.6426	0.5652
4	Exp 5	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.06], no warmup, n_trials=7, 85/15 split	lr=1.98e-05, wd=0.0098, warmup=0.0, epochs=5	head + tail	0.7102	0.6383	0.5154
5	Exp 7	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.06], warmup ∈ [0.06, 0.09], n_trials=7, 85/15 split	lr=1.55e-05, wd=0.0409, warmup=0.0744, epochs=5	head only	0.7048	0.6314	0.4758
6	Exp 3	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.06], warmup ∈ [0.06, 0.09], n_trials=10, 85/15 split	lr=1.56e-05, wd=0.0041, warmup=0.0813, epochs=5	head + tail	0.6904	0.6096	0.5301
7	Exp 6	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], wd ∈ [0.0, 0.06], warmup ∈ [0.06, 0.09], stopword removal, n_trials=7, 85/15 split	lr=1.93e-05, wd=0.0420, warmup=0.0657, epochs=4	head + tail	0.6674	0.5986	0.5108
8	Exp 4	lr ∈ [1.5e-5, 2e-5], batch=8, epochs ∈ [3,5], no weight decay , warmup ∈ [0.06, 0.09], n_trials=7, 85/15 split	lr=1.64e-05, wd=0.0, warmup=0.0791, epochs=5	head + tail	0.6913	0.5854	0.3465

wd == weight decay

Figure 4: DeBERTA trials ranked by performance.

DeBERTA Experiments — Ranked by Test F1 (weighted)

3.4.6. DeBERTA Experiments.

All experiments used the head + tail truncation strategy unless stated otherwise, as shown by Experiments 7 and 8, which instead used head only or tail only truncation and consistently achieved worse results, which suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity.

Experiment 1 — (Best Experiment)
The first systematic experiment for DeBERTA introduced optuna hyperparameter search across 5 trials with a broad search space: learning rate $\in$ [5e-6, 2e-5], weight decay $\in [0.0, 0.1]$, warmup ratio $\in [0.0, 0.1]$ and epochs $\in [3, 5]$. Batch size was fixed to 8 throughout all DeBERTA experiments due to GPU memory constraints. The best trial used lr=1.78e-05, weight decay=0.0499, warmup ratio=0.0794 and trained for 3 epochs achieving a validation F1 weighted of 0.6823 and a test F1 weighted of 0.6470, making it the best result across all DeBERTA experiments. The gap between validation and test F1 (0.6823 vs 0.6470) is the smallest across all experiments, confirming that the model generalized well to unseen data and was not overfitting.

Three factors combined to make Experiment 1 the best configuration overall. First the high weight decay of 0.0499, which is notably higher than BERT's best value of 0.0093, reflecting DeBERTA's larger number of parameters and its greater need for regularization. The Experiment 4 confirmed that weight decay is essential for DeBERTA on this task i.e. removing it entirely caused a complete collapse where the model predicted zero samples as Clear Non-Reply (F1=0.00) (see Table 11). This makes sense because weight decay acts as a regularizer by penalizing large weights, which prevents the model from memorizing the training data and improves generalization to unseen samples. Second a warmup ratio of 0.079 effectively stabilized early training, as confirmed by Experiment 5, which showed that removing warmup entirely lowered performance. The warmup ratio allows the learning rate to gradually increase from zero to its target value at the start of training, rather than starting with a large learning rate immediately, which prevents the model from making large unstable weight updates in the early epochs. Third, the head + tail truncation strategy which as confirmed by Experiments 7 and 8 which instead used head only or tail only truncation and consistently achieved worse results. This suggests that in political interviews both the beginning and the end of an answer carry important information for determining its clarity and that by preserving both parts we capture more of the information needed to determine the clarity label of a political answer.

Experiment 2 — Increasing the Number of Trials
This experiment used the same parameter search space as Experiment 1 but increased the number of trials from 5 to 10. This did not produce any significant improvement over Experiment 1 and therefore no meaningful conclusions could be drawn from it. For this reason Experiment 2 is skipped and we continue with Experiment 3.

Experiment 3 — Narrowed Search Space with More Trials
Based on the results of Experiment 1 the optuna search space was narrowed by restricting the learning rate to $\in$ [1.5e-5, 2e-5], weight decay to $\in [0.0, 0.06]$ and warmup

ratio to $\in [0.06, 0.09]$, while increasing the number of trials from 5 to 10 to explore the narrower space more thoroughly.

Despite the larger number of trials the test F1 weighted dropped from 0.6470 to 0.6096 compared to Experiment 1, while the validation F1 weighted increased to 0.6904, resulting in a larger gap between validation and test performance. This suggests that the model overfitted to the validation set and failed to generalize as well to unseen data. The main reason for this is that the best trial selected by optuna used a very low weight decay of 0.0041, which provided insufficient regularization compared to the weight decay of 0.0499 found in Experiment 1. Notably, Clear Reply performance dropped significantly with an F1 of only 0.33 compared to 0.43 in Experiment 1 (see Table 10), suggesting that insufficient regularization particularly hurts the model's ability to correctly identify the minority classes.

Experiment 4 — Removing Weight Decay
Weight decay was removed entirely to confirm whether it was the key factor behind Experiment 1 strong generalization. All other parameters kept the same ranges as Experiment 3. The result confirmed that weight decay is essential for DeBERTA on this task, the test F1 collapsed to 0.5854, the worst result across all DeBERTA experiments and most critically Clear Non-Reply collapsed entirely with F1=0.00, meaning the model predicted zero samples as this class, while Clear Reply also dropped to F1=0.27 compared to F1 = 0.33 in Experiment 3 (see Table 11). This is a direct consequence of removing weight decay, without it to penalize large weights, the model overfits to the dominant Ambivalent class and loses its ability to generalize to minority classes on unseen data. Comparing directly with Experiment 3 where weight decay was only 0.0041, even that very small amount of regularization was enough to prevent the complete collapse seen here. This result strongly confirms the finding from Experiment 1 that weight decay is essential when fine tuning DeBERTA on this task.

Experiment 5 — Removing Warmup Ratio
In this experiment warmup ratio was removed from the search space to investigate its importance for stable training. All other parameters remained the same as Experiment 3. The validation F1 weighted reached the highest value across all DeBERTA experiments at 0.7102, however the test F1 weighted dropped to 0.6383 compared to 0.6470 in Experiment 1, resulting in a gap between validation and test performance. This suggests that without warmup ratio the model overfits to the validation set despite achieving a high validation score and fails to generalize as well to unseen data. The warmup ratio plays an important role in stabilizing early training by gradually increasing the learning rate from zero to its target value, preventing large unstable weight updates in the early epochs. Without it the model starts with a large learning rate immediately, which can cause the model to make aggressive updates early in training that lead to solutions that perform well on the validation set but do not generalize well to unseen data. This confirms the finding from Experiment 1 that the combination of both weight decay and warmup ratio is necessary to achieve the best generalization performance on this task.

Experiment 6 — Stopword Removal
In this experiment a custom stopword removal step was added to the preprocessing pipeline removing common uninformative words before tokenization. All other parameters remained the same as Experiment 3. The test F1 weighted dropped to

0.5986 compared to 0.6096 in Experiment 3 and 0.6470 in Experiment 1, confirming that stopword removal hurts performance. This is because DeBERTA is a contextual model pretrained on complete unmodified text, so removing words before tokenization disrupts the natural language patterns DeBERTA learned during pretraining and therefore hurts performance. DeBERTA already handles context and word importance internally through its attention mechanism, so manual removal of words before tokenization is not only unnecessary but actively harmful as it removes tokens that DeBERTA relies on to understand the meaning of the input. This confirms that manual preprocessing steps are generally not beneficial when fine tuning transformer models, as confirmed by the same finding in the BERT and DistilBERT experiments where stopword removal also consistently hurt performance.

Experiment 7 — Head Only Truncation
The head + tail truncation strategy was replaced with head only truncation to investigate whether the beginning of a political answer is more informative than its end for clarity classification. All other parameters remained the same as Experiment 3. The test F1 weighted dropped from 0.6470 in Experiment 1 to 0.6314, confirming that discarding the tail of the answer hurts performance and that the model did not generalize as well to unseen data. This suggests that the conclusion of a political answer also carries important clarity signals, for example a politician may start ambiguously but give a clearer or more evasive conclusion at the end. By keeping only the beginning of the answer the model loses access to this information.

Experiment 8 — Tail Only Truncation
The head + tail truncation strategy was replaced with tail only truncation to investigate whether the end of a political answer is more informative than its beginning for clarity classification. All other parameters remained the same as Experiment 3. The test F1 weighted dropped from 0.6470 in Experiment 1 to 0.6426, confirming that discarding the head of the answer hurts performance. Interestingly, tail only truncation performed better than head only truncation (0.6426 vs 0.6314), suggesting that the end of a political answer carries slightly more clarity signals than its beginning. However both strategies are clearly inferior to head + tail truncation. Comparing all three truncation strategies head only (0.6314), tail only (0.6426) and head + tail (0.6470) confirms that the head + tail strategy is clearly superior, suggesting that in political interviews the most relevant information for determining clarity is distributed across both the beginning and the end of an answer, and that preserving both parts leads to better classification performance.

Experiment 9 — Smaller Validation Split (90/10)
The validation split was reduced from 15% to 10% to give the model more training data, while all other parameters remained the same as Experiment 3. This experiment achieved the highest test F1 weighted of 0.6487 across all DeBERTA experiments, surpassing Experiment 1 (0.6470). However the gap between the two is extremely small (0.0017), which means the improvement is not meaningful and the two experiments should be considered essentially equivalent in performance. It is also worth noting that the same change produced the opposite effect for both BERT and DistilBERT, where reducing the validation split to 10% consistently led to worse test F1 results compared to the 85/15 split. This difference can be explained by the fact that DeBERTA is a significantly larger and more powerful model than BERT and DistilBERT, with more

parameters and a stronger ability to generalize. This means that DeBERTA is less sensitive to the size of the validation set used for model selection. BERT and DistilBERT on the other hand are smaller and simpler models that are more prone to overfitting on this relatively small dataset and therefore benefit more from having a larger validation set to guide training and prevent the model from memorizing the training data.

Experiment 1 (Best)

Class	Precision	Recall	F1
Clear Non-Reply	0.37	0.70	0.48
Ambivalent	0.76	0.74	0.75
Clear Reply	0.48	0.39	0.43

Table 9: Per class results for DeBERTA Experiment 1 (best experiment)

Experiment 3

Class	Precision	Recall	F1
Clear Non-Reply	0.45	0.65	0.54
Ambivalent	0.73	0.72	0.73
Clear Reply	0.34	0.32	0.33

Table 10: Per class results for DeBERTA Experiment 3

Experiment 4 (No Weight Decay)

Class	Precision	Recall	F1
Clear Non-Reply	0.00	0.00	0.00
Ambivalent	0.70	0.86	0.77
Clear Reply	0.32	0.23	0.27

Table 11: Per class results for DeBERTA Experiment 4 (no weight decay)

4. Evaluation

4.1. Metrics

Model performance was evaluated using four standard classification metrics: accuracy, precision, recall and F1-score.

> Accuracy measures the overall proportion of correctly classified samples.

> Precision measures how many of the samples predicted as a given class were actually that class.

> Recall measures how many of the actual samples of a given class were correctly identified by the model.

F1-score is the harmonic mean of precision and recall, providing a single balanced metric that accounts for both.

Since the dataset is significantly imbalanced across the three clarity labels (Ambivalent: 59.2%, Clear Reply: 30.5%, Clear Non-Reply: 10.3%), accuracy alone would be misleading, as a model that always predicts the majority class would still achieve a high accuracy score without actually learning to distinguish between classes. For this reason weighted F1 was chosen as the primary evaluation metric, as it is the official metric used in the Kaggle competition for this assignment. It computes the F1-score for each class and averages them proportionally to their support in the dataset. Macro F1 is also reported as an additional metric, as it treats all three classes equally regardless of their size, giving equal importance to minority classes.

All metrics were computed using scikit-learn's classification_report function, which provides per-class precision, recall and F1-score.

4.2. Learning Curve

4.2.1. Training & Validation Loss per epoch — DistilBERT (Experiment 3 Best One).



Figure 5: Training & Validation Loss per Epoch — DistilBERT (Experiment 3 Best One)

The training loss decreases steadily across all five epochs, dropping from 1.06 at epoch 1 to 0.57 at epoch 5. The validation loss also decreases overall, from 1.00 at epoch 1 to 0.84 at epoch 5, however it shows a slight increase at epoch 3 (0.88) before continuing to decrease. From epoch 3 onwards the training loss continues to drop significantly while the validation loss stabilizes and decreases only slightly, creating a growing gap between the two curves. This divergence suggests that mild overfitting begins after epoch 3, where the model starts memorizing the training data rather than generalizing further.

4.2.2. Validation F1 Scores per Epoch — DistilBERT (Experiment 3 Best One).

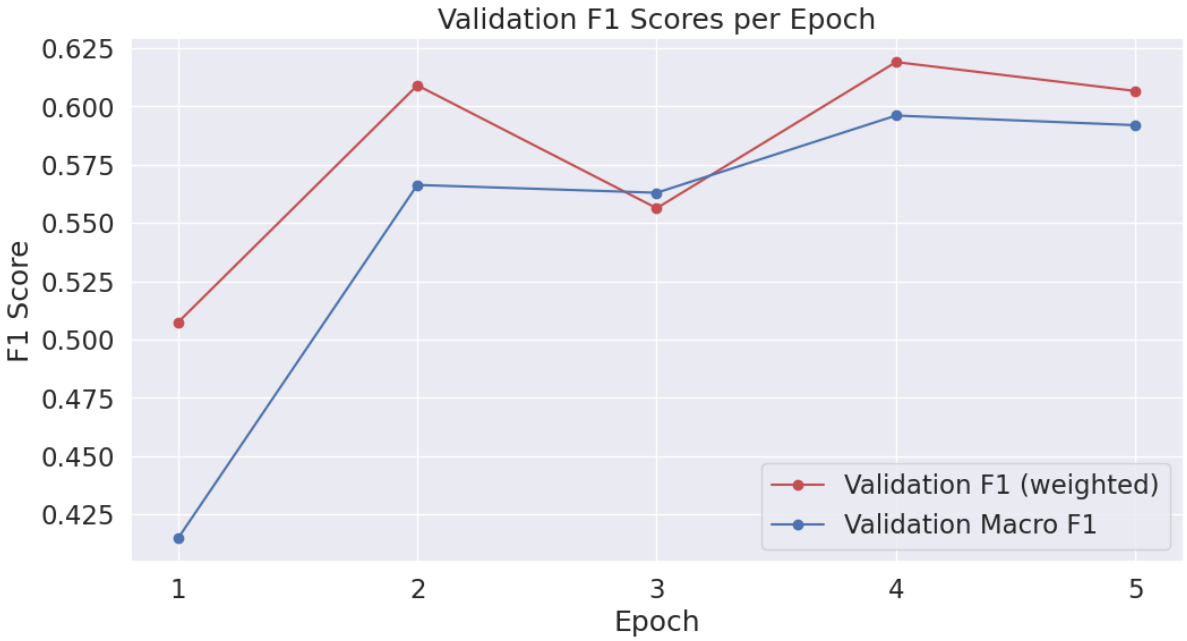


Figure 6: Validation F1 Scores per Epoch — DistilBERT (Experiment 3 Best One)

The validation F1 weighted starts at 0.51 at epoch 1, jumps to 0.61 at epoch 2, drops back to 0.56 at epoch 3, then reaches its peak of 0.62 at epoch 4 before slightly dropping to 0.61 at epoch 5. The Macro F1 follows a similar pattern, starting very low at 0.41 at epoch 1 and gradually improving to 0.60 at epoch 4. The best model was saved at epoch 4 with a validation F1 weighted of 0.6581, which is the best result achieved across all DistilBERT experiments. The drop at epoch 3 before recovering at epoch 4 suggests some training instability during the middle epochs, which is consistent with the warmup ratio gradually ramping up the learning rate during the early epochs.

4.2.3. Training & Validation Loss per Epoch — BERT (Experiment 4 Best One).



Figure 7: Training & Validation Loss per Epoch — BERT (Experiment 4 Best One)

The training loss decreases steadily across all four epochs, dropping from 1.05 at epoch 1 to 0.59 at epoch 4. The validation loss however shows a less stable pattern it decreases from 1.00 at epoch 1 to 0.86 at epoch 2, then slightly increases at epoch 3 (0.87) and increases more noticeably at epoch 4 (0.91). This divergence between training and validation loss from epoch 2 onwards indicates that the model begins to overfit after epoch 2, where the training loss keeps decreasing while the validation loss starts rising. Despite this, the weight decay regularization helped keep the gap between validation F1 weighted (0.6694) and test F1 weighted (0.6500) small, confirming that the model still generalizes reasonably well to unseen data.

4.2.4. Validation F1 Scores per Epoch — BERT (Experiment 4 Best One).

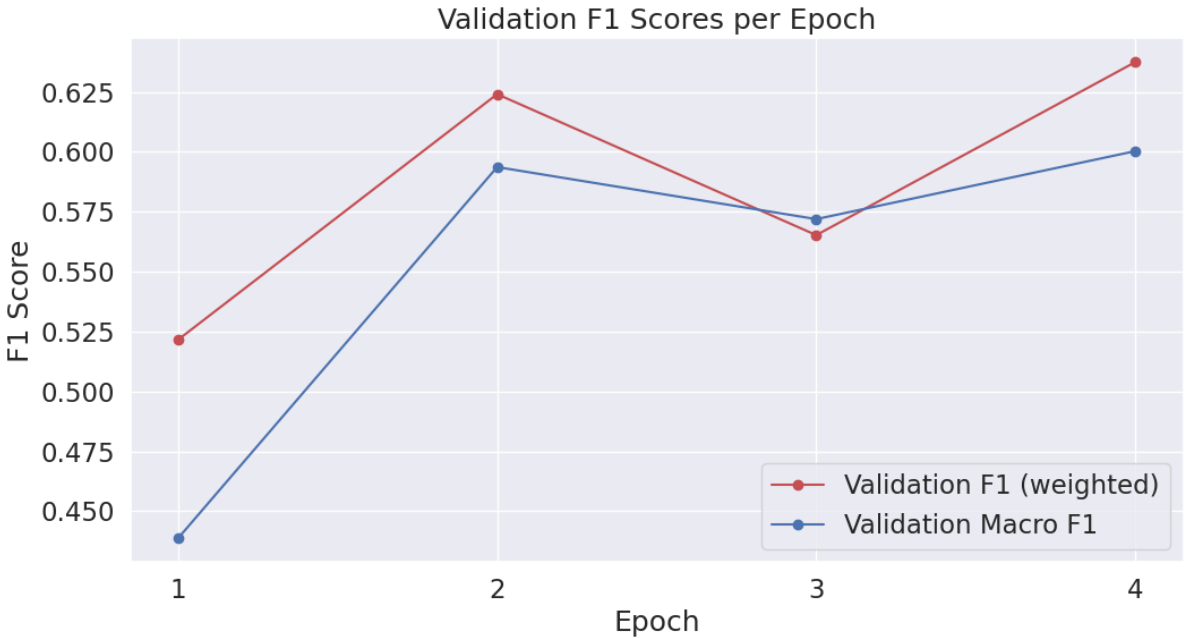


Figure 8: Validation F1 Scores per Epoch — BERT (Experiment 4 Best One)

The validation F1 weighted starts at 0.52 at epoch 1, jumps to 0.62 at epoch 2, drops back to 0.57 at epoch 3, then reaches its peak of 0.64 at epoch 4. The Macro F1 follows a similar pattern, starting at 0.44 at epoch 1 and gradually improving to 0.60 at epoch 4. Both curves show a dip at epoch 3 before recovering and reaching their best values at epoch 4. The best model was saved at epoch 4 with a validation F1 weighted of 0.6694, which is the best result achieved across all BERT experiments.

4.2.5. Training & Validation Loss per Epoch — DeBERTA (Experiment 1 Best One).



Figure 9: Training & Validation Loss per Epoch — DeBERTA (Experiment 1 Best One)

The training loss decreases steadily across all three epochs, dropping from 1.04 at epoch 1 to 0.70 at epoch 3. The validation loss also decreases overall, from 1.02 at epoch 1 to 0.73 at epoch 3, however it shows a notable drop at epoch 2 (0.75) before continuing to decrease slightly at epoch 3 (0.73). Notably the validation loss is lower than the training loss at epochs 1 and 2, however at epoch 3 the training loss drops below the validation loss (0.70 vs 0.73), suggesting that the model begins to fit the training data more closely in the final epoch while still maintaining good generalization performance. This behavior is consistent with the strong weight decay of 0.0499 acting as a regularizer, preventing the model from memorizing the training data and keeping the validation loss close to the training loss throughout training. The small gap between training loss (0.70) and validation loss (0.73) at epoch 3 confirms that the model generalized well to unseen data, which is also reflected in the small gap between validation F1 weighted (0.6823) and test F1 weighted (0.6470).

4.2.6. Validation F1 Scores per Epoch — DeBERTA (Experiment 1 Best One).

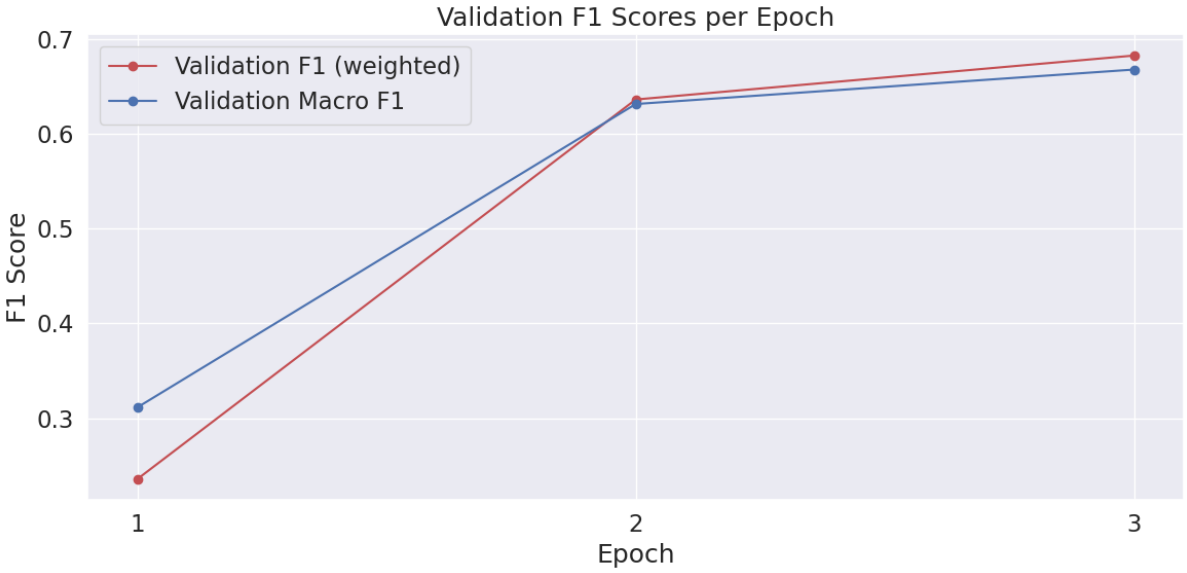


Figure 10: Validation F1 Scores per Epoch — DeBERTA (Experiment 1 Best One)

The validation F1 weighted starts very low at 0.24 at epoch 1, then jumps significantly to 0.64 at epoch 2, and continues to improve to its peak of 0.68 at epoch 3. The Macro F1 follows a very similar pattern, starting at 0.31 at epoch 1, jumping to 0.63 at epoch 2, and reaching 0.67 at epoch 3. Both curves show a consistent and steep improvement across all three epochs with no drops or instability, suggesting that the model learned steadily and reliably throughout training. The best model was saved at epoch 3 with a validation F1 weighted of 0.6823, which is the best result achieved across all DeBERTA experiments. Unlike BERT and DistilBERT which both showed a temporary dip at epoch 3 before recovering, DeBERTA shows a smooth monotonic improvement across all three epochs.

4.2.7. Comparison BERT vs DistilBERT vs DeBERTA.

BERT and DistilBERT both show a similar training pattern i.e. the training loss decreases steadily across all epochs while the validation loss stabilizes or slightly increases in the later epochs, indicating mild overfitting in both cases. However BERT shows a more pronounced divergence between training and validation loss from epoch 2 onwards, while DistilBERT maintains a smaller gap throughout training, suggesting that DistilBERT is slightly more stable during training. Despite this, BERT achieves a better final validation F1 weighted (0.6694 vs 0.6581) and a better test F1 weighted (0.6500 vs 0.6219), confirming that BERT's larger capacity of 110M parameters allows it to learn more complex patterns from the question-answer pairs compared to DistilBERT's 66M parameters. This is expected given that DistilBERT was designed to be a smaller and faster version of BERT, being 40% smaller and 60% faster, trading some predictive performance for efficiency. Both models show a temporary drop in validation F1 at epoch 3 before recovering and reaching their best performance at epoch 4, suggesting some training instability during the middle epochs. For DistilBERT this behavior is consistent with the warmup ratio gradually ramping up the learning rate during the early epochs.

DeBERTA on the other hand shows a smooth monotonic improvement in validation F1 across all three epochs with no instability or dips, achieving its best validation F1 weighted of 0.6823 at epoch 3. The validation loss remains close to the training loss throughout training, confirming that the strong weight decay of 0.0499 effectively prevented overfitting. However despite achieving the highest validation F1 weighted among all three models, DeBERTA test F1 weighted of 0.6470 is slightly lower than BERT 0.6500, suggesting that BERT generalized slightly better to the unseen test data. This is a surprising result given that DeBERTA is a significantly larger and more powerful model with 184M parameters compared to BERT 110M parameters and this might be related to the fact that DeBERTA was trained for only 3 epochs compared to 4 epochs for BERT, which may have limited the model from reaching its full potential on this task.

4.3. Confusion Matrices

4.3.1. DistilBERT (Experiment 3 Best One).

As shown in Figure 11, the model performs best on the Ambivalent class, correctly classifying 151 out of 206 samples (73.3%), while misclassifying 15 as Clear Non-Reply and 40 as Clear Reply. This is expected given that Ambivalent is the dominant class with 59.2% of the samples.

The model struggles significantly with the two minority classes. For Clear Non-Reply, only 13 out of 23 samples (56.5%) were correctly classified, with 7 misclassified as Ambivalent and 3 as Clear Reply. For Clear Reply, only 31 out of 79 samples (39.2%) were correctly classified, with 45 misclassified as Ambivalent and 3 as Clear Non-Reply.

The most common error pattern across both minority classes is misclassification toward Ambivalent, which is consistent with the class imbalance present in the dataset.

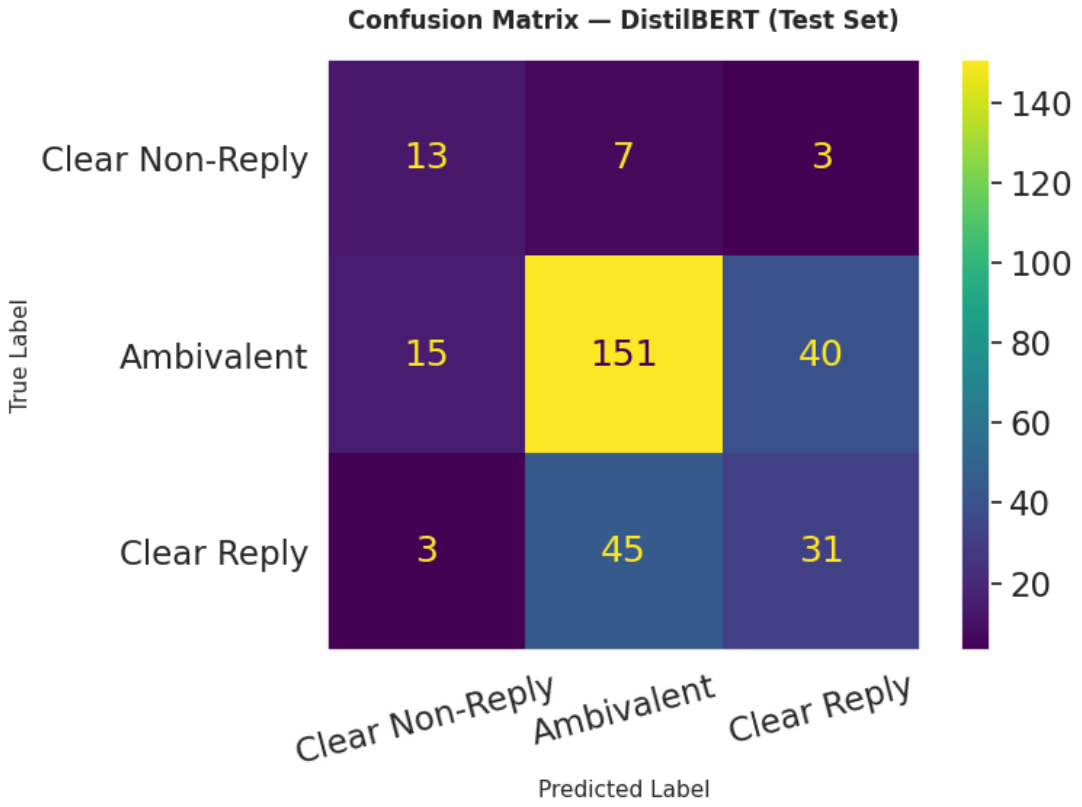


Figure 11: Confusion Matrix — DistilBERT (Test Set)

4.3.2. BERT (Experiment 4 Best One).
As shown in Figure 12, the model performs best on the Ambivalent class, correctly classifying 164 out of 206 samples (79.6%), while misclassifying 18 as Clear Non-Reply and 24 as Clear Reply. This is expected given that Ambivalent is the dominant class with 59.2% of the samples.

The model struggles significantly with the two minority classes. For Clear Non-Reply, only 12 out of 23 samples (52.2%) were correctly classified, with 8 misclassified as Ambivalent and 3 as Clear Reply. For Clear Reply, only 27 out of 79 samples (34.2%) were correctly classified, with 48 misclassified as Ambivalent and 4 as Clear Non-Reply.

The most common error pattern across both minority classes is misclassification toward Ambivalent, which is consistent with the class imbalance present in the dataset.

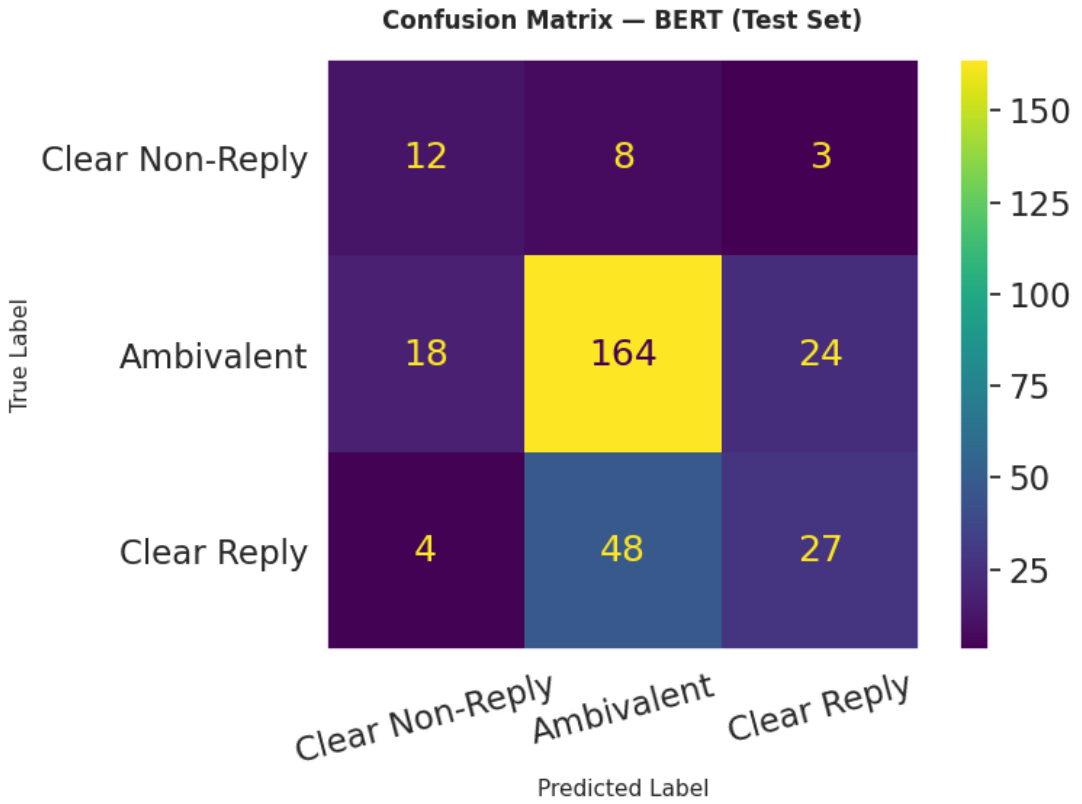


Figure 12: Confusion Matrix — BERT (Test Set)

4.3.3. DeBERTA (Experiment 1 Best One).

As shown in Figure 13, the model performs best on the Ambivalent class, correctly classifying 152 out of 206 samples (73.8%), while misclassifying 18 as Clear Non-Reply and 36 as Clear Reply. This is expected given that Ambivalent is the dominant class with 59.2% of the samples.

The model struggles significantly with the two minority classes. For Clear Non-Reply, only 16 out of 23 samples (69.6%) were correctly classified, with 5 misclassified as Ambivalent and 2 as Clear Reply. For Clear Reply, only 31 out of 79 samples (39.2%) were correctly classified, with 45 misclassified as Ambivalent and 3 as Clear Non-Reply.

The most common error pattern across both minority classes is misclassification toward Ambivalent, which is consistent with the class imbalance present in the dataset.

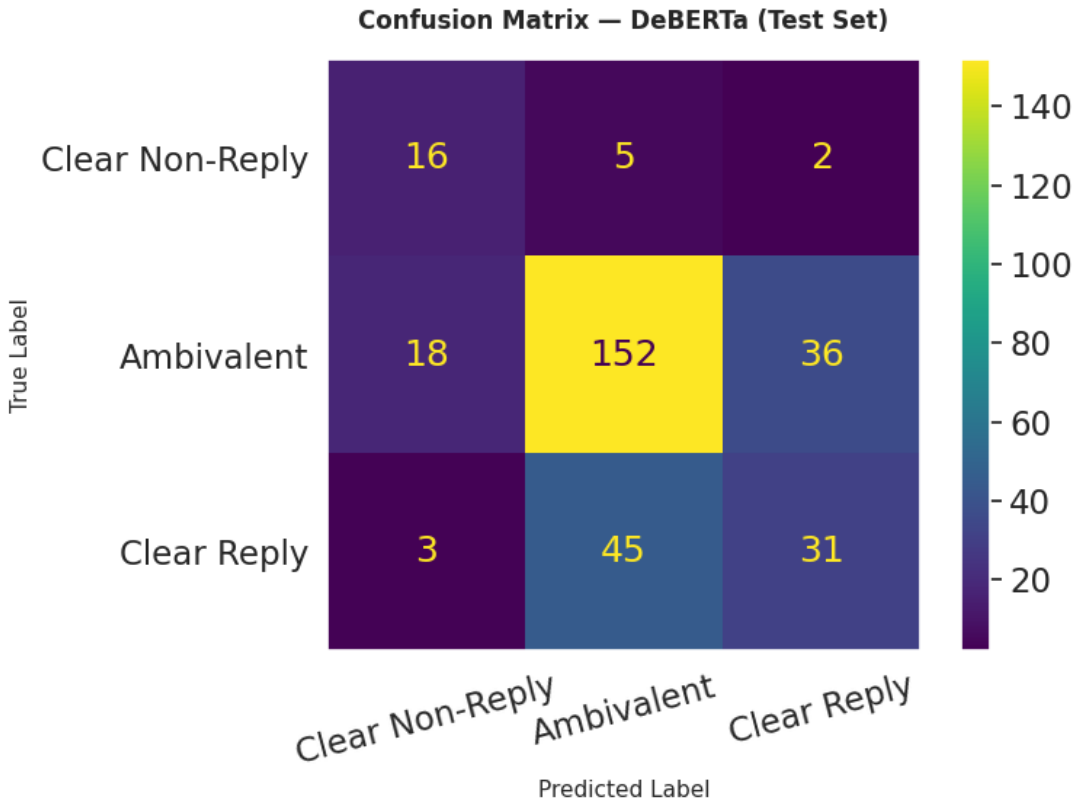


Figure 13: Confusion Matrix — DeBERTA (Test Set)

Comparison DistilBERT vs BERT vs DeBERTA

The table below summarizes the per-class accuracy, Test F1 weighted and Test Macro F1 for all three models:

Class	DistilBERT	BERT	DeBERTa
Ambivalent	151/206 = 73.3%	164/206 = 79.6%	152/206 = 73.8%
Clear Non-Reply	13/23 = 56.5%	12/23 = 52.2%	16/23 = 69.6%
Clear Reply	31/79 = 39.2%	27/79 = 34.2%	31/79 = 39.2%
Test F1 weighted	0.6219	**0.6500**	0.6470
Test Macro F1	0.5111	**0.5809**	0.5551

Table 12: Per-class accuracy comparison: DistilBERT vs BERT vs DeBERTA (Test Set)

All three models share the same error pattern i.e. the majority of misclassifications from both minority classes go toward Ambivalent, confirming that the dominant class attracts most incorrect predictions regardless of model size or architecture. As shown in Table 12, BERT outperforms both DistilBERT and DeBERTA on the Ambivalent class (79.6% vs 73.3% and 73.8%) and on the overall metrics, Test F1 weighted (0.6500 vs 0.6219 and 0.6470) and Test Macro F1 (0.5809 vs 0.5111 and 0.5551). However DeBERTA performs best on the Clear Non-Reply class (69.6% vs 56.5% and 52.2%), while DistilBERT and DeBERTA perform equally on Clear Reply (39.2% vs 34.2%).

5. Results and Overall Analysis

5.1. Results Table

Table 13 presents the overall metrics for all three models on the test set and Tables 14, 15 and 16 present the per-class precision, recall and F1 scores. BERT achieves the best overall performance with a test F1 weighted of 0.6500 and a Macro F1 of 0.5809, outperforming both DistilBERT and DeBERTA on both metrics.

Model	Accuracy	F1 Weighted	Macro F1
BERT	0.6461	**0.6500**	**0.5809**
DistilBERT	0.6526	0.6219	0.5111
DeBERTA	0.6461	0.6470	0.5551

Table 13: Overall test set results for all three models

BERT (Experiment 4)

Class	Precision	Recall	F1
Clear Non-Reply	0.54	0.61	0.57
Ambivalent	0.76	0.73	0.74
Clear Reply	0.41	0.44	0.43

Table 14: Per-class results for BERT on the test set

DistilBERT (Experiment 3)

Class	Precision	Recall	F1
Clear Non-Reply	0.42	0.61	0.50
Ambivalent	0.72	0.83	0.77
Clear Reply	0.41	0.19	0.26

Table 15: Per-class results for DistilBERT on the test set

DeBERTA (Experiment 1)

Class	Precision	Recall	F1
Clear Non-Reply	0.37	0.70	0.48
Ambivalent	0.76	0.74	0.75
Clear Reply	0.48	0.39	0.43

Table 16: Per-class results for DeBERTA on the test set

5.2. Best Model

DistilBERT Best Configuration (Experiment 3)
The best configuration for DistilBERT was found in Experiment 3 with the following parameters: learning rate 1.95e-05, weight decay 0.0571, warmup ratio 0.0969, epochs 5

and batch size 16 using the head + tail truncation strategy. This configuration achieved a test F1 weighted of 0.6219 and a Macro F1 of 0.5111. The key factors that made this the best configuration were the combination of warmup ratio which stabilized early training updates, high weight decay which prevented overfitting and the head + tail truncation strategy which preserved the most important parts of both the beginning and end of political answers.

BERT Best Configuration (Experiment 4)
The best configuration for BERT was found in Experiment 4 with the following parameters: learning rate 2.99e-05, weight decay 0.0093, epochs 4 and batch size 16 using the head + tail truncation strategy. This configuration achieved a test F1 weighted of 0.6500 and a Macro F1 of 0.5809. The key factor that made this the best configuration for BERT was the addition of weight decay to the optuna search space, which significantly reduced the overfitting observed in Experiment 3 for this model where no weight decay was used, combined with the head + tail truncation strategy which preserved the most informative parts of the political answers.

Notably for BERT weight decay had a much larger impact than for DistilBERT, improving the test F1 weighted from 0.5636 to 0.6500. This can be attributed to BERT's larger number of parameters, with 110M parameters BERT has a much higher capacity to overfit the training data and therefore benefits more from regularization through weight decay to prevent overfitting. For DistilBERT the most impactful change was instead the addition of warmup ratio, which gradually increases the learning rate from zero to its target value at the start of training. This is particularly beneficial for smaller models like DistilBERT because they have fewer parameters to adjust and are more sensitive to large weight updates in the early epochs, by starting with a very small learning rate and gradually increasing it, the model avoids making large unstable updates before the weights have had a chance to settle into a good region of the parameter space.

DeBERTA Best Configuration (Experiment 1)
The best configuration for DeBERTA was found in Experiment 1 with the following parameters: learning rate 1.78e-05, weight decay 0.0499, warmup ratio 0.0794, epochs 3 and batch size 8 using the head + tail truncation strategy. This configuration achieved a test F1 weighted of 0.6470 and a Macro F1 of 0.5551. The key factors that made this the best configuration were the combination of a high weight decay of 0.0499 which is notably much higher than BERT's value of 0.0093 for weight decay, confirming that DeBERTA with 184M parameters requires stronger regularization than BERT with 110M parameters to prevent overfitting on this relatively small dataset, the warmup ratio which stabilized early training updates and the head + tail truncation strategy which preserved the most informative parts of the political answers.

Overall Best Model
BERT (Experiment 4) is the best overall model, achieving the highest test F1 weighted (0.6500) and Macro F1 (0.5809) across all three models. This is a surprising result given that DeBERTA is a significantly larger and more powerful model with 184M parameters compared to BERT 110M parameters. However DeBERTA test F1 weighted of 0.6470 is only slightly lower than BERT 0.6500, suggesting that the two models perform essentially equivalently on this task and this might be related to the fact that DeBERTA was trained for only 3 epochs compared to 4 epochs for BERT, which may have limited

the model from reaching its full potential on this task. DistilBERT achieves the lowest performance with a test F1 weighted of 0.6219, which is expected given that it was designed to be a smaller and faster version of BERT, being 40% smaller and 60% faster, trading some predictive performance for efficiency.

6. Error Analysis

6.1. Which class is hardest to predict?

Clear Reply is consistently the hardest class to predict across all three models. BERT achieved an F1 of 0.43, DeBERTA achieved an F1 of 0.43 and DistilBERT achieved an F1 of only 0.26 on this class, making it the lowest scoring class for all three models. Ambivalent is the easiest class to predict with an F1 of 0.74 for BERT, 0.75 for DeBERTA and 0.77 for DistilBERT, while Clear Non-Reply falls in between with an F1 of 0.57 for BERT, 0.48 for DeBERTa and 0.50 for DistilBERT.

The difficulty of Clear Reply can be attributed to its low representation in the training data (30.5% compared to 59.2% for Ambivalent) and its linguistic similarity to Ambivalent answers, making it hard for the model to distinguish between the two classes. This is confirmed by the confusion matrices where the vast majority of Clear Reply misclassifications go toward Ambivalent specifically 48 out of 52 misclassified Clear Reply samples for BERT, 45 out of 48 for DistilBERT and 45 out of 48 for DeBERTa were predicted as Ambivalent (see Figures 12, 11 and 13).

6.2. Do stronger or larger models fail differently from smaller baseline models?

All three models share the same general error pattern i.e. the majority of misclassifications from both minority classes go toward Ambivalent, confirming that the dominant class attracts most incorrect predictions regardless of model size or architecture.

However the three models fail differently in terms of where they are stronger and weaker. BERT performs significantly better on the Ambivalent class, correctly classifying 164 out of 206 samples (79.6%) compared to 152 out of 206 (73.8%) for DeBERTA and 151 out of 206 (73.3%) for DistilBERT. This suggests that BERT captures the complex linguistic patterns of the dominant Ambivalent class better than both DeBERTA and DistilBERT, which is a somewhat unexpected finding given that DeBERTA is a significantly larger and more powerful model than BERT with 184M parameters compared to 110M and might be partly related to the fact that BERT trained for 4 epochs compared to only 3 epochs for DeBERTA, giving it more time to learn the patterns of this dominant class. On the other hand DeBERTA performs best on the Clear Non-Reply class, correctly classifying 16 out of 23 samples (69.6%) compared to 13 out of 23 (56.5%) for DistilBERT and 12 out of 23 (52.2%) for BERT. For Clear Reply, DistilBERT and DeBERTA perform equally (39.2% each), both outperforming BERT which achieved only 34.2%.

Overall BERT achieves the best performance on both metrics: Test F1 weighted (0.6500 vs 0.6470 and 0.6219) and Test Macro F1 (0.5809 vs 0.5551 and 0.5111), confirming that despite being smaller than DeBERTA, BERT generalizes slightly better to unseen data on this task, which might be partly related to the fact that BERT was trained for 4

epochs compared to only 3 epochs for DeBERTA. But notably DeBERTA outperforms BERT on both minority classes, Clear Non-Reply (69.6% vs 52.2%) and Clear Reply (39.2% vs 34.2%), suggesting that its larger and more powerful architecture gives it an advantage on the hardest and most underrepresented classes in the dataset even if it was trained for fewer epochs than BERT.

6.3. What is the fundamental source of error: what do models fail to understand in question–answer interactions?

There are multiple fundamental sources of error that contribute to model failures on this task.

Class Imbalance
The dataset is significantly imbalanced with Ambivalent dominating at 59.2% of the training data. Although class-weighted loss was used during training to give higher penalty to minority class errors and helped improve minority class performance, the bias toward predicting Ambivalent still persists as evidenced by the confusion matrices where the majority of misclassifications from both minority classes go toward Ambivalent. This suggests that class weights alone are not sufficient to fully overcome such a class imbalance.

Long Answer Truncation
Approximately 30% of the combined question-answer token sequences in this dataset exceed the 512 token limit supported by all three models and must therefore be truncated. Even with the head + tail truncation strategy which preserves both the beginning and the end of the answer, a significant portion of the answer content is lost. The model may therefore miss important clarity signals that appear in a long political answer, which could lead to misclassifications.

Political Language Complexity
Political speech is inherently complex and difficult to classify, making it very difficult for the model to determine whether an answer is a Clear Reply, Ambivalent or a Clear Non-Reply. This challenge is further worsened by the fact that BERT and DistilBERT were pretrained on BookCorpus and Wikipedia, while DeBERTA was pretrained on Wikipedia and BookCorpus as well as additional web crawl data, none of which contain political interview data.

Pretraining on Dialogue and QA Data
This task requires understanding the relationship between a question asked by a journalist and an answer given by a president. However none of the three models were pretrained on question-answer data. BERT and DistilBERT were pretrained on BookCorpus and Wikipedia, while DeBERTA was pretrained on Wikipedia and BookCorpus as well as additional web crawl data, none of which contain interview data. As a result the models may struggle to fully capture the linguistic complexity of question-answer interactions that is necessary for determining whether the answer actually addresses the question, since they were pretrained exclusively on encyclopedic and literary text rather than on interview data.

6.4. What characteristics would an ideal model need in order to improve performance on this task?

Based on the analysis of model failures, an ideal model would need several key characteristics to improve performance on this task.

Longer Context Window
All three models are limited to 512 tokens while approximately 30% of the combined question-answer token sequences in the dataset exceed this limit. An ideal model would be able to process the full sequence without truncation, preserving all the clarity signals present in the complete answer.

Better Handling of Class Imbalance
Although class weighted loss was used during training to address the class imbalance and helped improve minority class performance, an ideal model would additionally benefit from further techniques to better learn the patterns of the minority classes and reduce the bias toward the dominant Ambivalent class.

Understanding of Political Language
BERT and DistilBERT were pretrained on BookCorpus and Wikipedia, while DeBERTA was pretrained on Wikipedia and BookCorpus as well as additional web crawl data, none of which contain political interview data. An ideal model would benefit from pretraining on political interview data, as political language has different patterns that general pretraining on encyclopedic and literary text may not capture well.

Pretraining on Dialogue and QA Data
Since this task involves understanding the relationship between a question asked by a journalist and an answer given by a president, an ideal model would benefit from pretraining on question-answer data. Models pretrained on such data would better understand the interaction between a question and its answer, which is fundamental for determining whether the answer actually addresses the question.

6.5. Summary of Attempted Changes

A detailed description of all the changes attempted to improve all three models, whether these changes improved or hurt performance, and the reasons behind each outcome are discussed in detail in Section 3.4. Here we provide a brief summary of the most important findings.

Changes that improved performance
The most beneficial change for BERT was the addition of weight decay to the optuna search space, which significantly reduced overfitting and improved the test F1 weighted from 0.5636 to 0.6500. This can be attributed to BERT's larger number of parameters i.e. with 110M parameters BERT has a much higher capacity to overfit the training data and therefore benefits more from weight decay regularization to prevent overfitting.

For DistilBERT the most beneficial change was the addition of warmup ratio, which improved the test F1 weighted from 0.5583 to 0.6219. This is particularly beneficial for smaller models like DistilBERT because they have fewer parameters to adjust and are more sensitive to large weight updates in the early epochs, by starting with a very small learning rate and gradually increasing it, the model avoids making large unstable

updates before the weights have a chance to settle into a good region of the parameter space.

For DeBERTA weight decay proved to be the most critical factor, as confirmed by Experiment 4 which showed that removing it entirely caused the model to completely fail to predict the Clear Non-Reply class with an F1 of 0.00, making it the most impactful hyperparameter for DeBERTA on this task. This can be attributed to DeBERTA even larger number of parameters i.e. with 184M parameters DeBERTA has an even higher capacity to overfit the training data compared to BERT's 110M parameters, which is reflected in the best weight decay value found by optuna search where DeBERTA required a higher weight decay of 0.0499, confirming that DeBERTA requires stronger regularization to prevent overfitting on this relatively small dataset. In DeBERTA the warmup ratio also proved important, as confirmed by Experiment 5 which showed that removing it led to a drop in test F1 weighted, though its impact was less severe than weight decay.

For all three models the head + tail truncation strategy consistently achieved better results than head only or tail only truncation, and the 85/15 validation split proved more effective than the 90/10 split for BERT and DistilBERT, while for DeBERTA both splits produced essentially equivalent results.

Changes that hurt performance
Stopword removal was the most harmful change for BERT and DistilBERT. For DistilBERT it caused Clear Reply to completely collapse with a recall and F1 of 0.00. For DeBERTA it produced the second worst test F1 weighted across all DeBERTA experiments (0.5986), confirming that transformer models handle text preprocessing internally and do not benefit from manual stopword removal. For DeBERTA the removal of weight decay had a catastrophic impact, causing the model to completely collapse to predict the Clear Non-Reply class with an F1 of 0.00 and producing the worst test F1 weighted across all DeBERTA experiments (0.5854), confirming that weight decay is essential for DeBERTA on this task. Both head only and tail only truncation strategies consistently underperformed compared to head + tail across all three models. Reducing the validation split to 90/10 consistently hurt performance for BERT and DistilBERT, while for DeBERTa both splits produced essentially equivalent results.

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